YIFEI SUN

Principal Software Engineer · API & Developer Platform Architecture · 13 Years

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SUMMARY

Principal-level backend & platform engineer with 13 years of experience designing production APIs, distributed systems, and platform infrastructure at scale. Recent work includes building an MCP server on the OpenAI Apps SDK — designing API contracts and serving infrastructure for ChatGPT integration — and architecting a high-traffic API modernization that cut p99 latency 5× while reaching 99.9% availability. Track record of owning major platform capabilities end-to-end, driving technical direction across 10+ teams, and shipping complex 0→1 systems from inception through production with a high bar for API design, reliability, and developer experience.

LEADERSHIP HIGHLIGHTS

- Designed and shipped production APIs and platform services powering millions of daily requests across multiple consumer types
- Built on OpenAI Apps SDK (MCP server) — firsthand understanding of the developer platform experience
- Led multi-team platform migrations spanning 10+ engineering teams across 3 departments
- Mentored senior engineers toward Principal-level scope; drove cross-org architectural standards via RFCs

Technologies: Java, Spring Boot, Python, Node.js · AWS, Kubernetes, Docker, Kafka, Redis, PostgreSQL · GraphQL (Netflix DGS), REST API Design, Microservices, Event-Driven, CQRS · OpenAI Apps SDK, MCP Protocol, LLM Integration · Datadog, Grafana · Keycloak, SSO, OAuth 2.0

INDEPENDENT PROJECT

GenAI Platform — Inference Infrastructure & Evaluation System

- Architected a platform with hybrid rules + multi-path LLM inference (fast vs. deep), configurable routing across multiple LLM providers, and a policy overlay — producing explainable real-time recommendations over heterogeneous data sources.
- Built an evaluation and experimentation system ("Prompt Lab") for end-to-end simulation, offline backtesting, and prompt/parameter search (grid & Bayesian optimization) with robust observability and multi-provider LLM support.
- Designed a configurable policy framework with DB-backed configs, environment-specific overrides, and a scheduler-driven tuning loop that sweeps parameters and publishes promotable config versions — enabling safe, iterative system improvement.

EXPERIENCE

CarGurus — Principal Software Engineer / Technical Lead Oct 2019 – Present

OpenAI Apps SDK MCP Server — Developer Platform Integration (0→1)

- Sole backend engineer designing and implementing an MCP server on the OpenAI Apps SDK, exposing CarGurus capabilities into ChatGPT — defining the API contracts, tool schema, serving infrastructure, and request/response interfaces with care for developer experience.
- Delivered a production-grade implementation under ambiguity; iterated rapidly with PMs and directors on API surface design and behavior; designed for safe, incremental rollout alongside existing APIs and consumers.
- Gained firsthand experience as a builder on OpenAI's developer platform — understanding integration pain points, SDK ergonomics, and API design tradeoffs from the consumer side.

High-Traffic API Modernization — Core Platform Infrastructure

- Led 3 principal/senior engineers and partnered with ~10 teams across 3 departments to redesign the core API behind CarGurus' most-visited page (~8,000 rpm, millions of daily impressions); new implementation handled 50% of traffic during A/B rollout.
- Introduced a new orchestration service and API boundary, consolidating 12 downstream integrations into 5 well-defined services with clear contracts, reducing fan-out and simplifying failure modes.
- Produced initial architecture in ~2 weeks; shipped the first production milestone in 6 months and completed the overall modernization ahead of schedule.

p99: 250 ms → 50 ms (5×) **Availability: 98% → 99.9%** **Deploy: 1 hr → 3 min** (20×)

- Defined API design standards for new microservices; built a parity-testing framework validating full payload and data-integrity against the legacy stack; removed years of hidden domain logic with clear documentation.

GraphQL Federation — Unified API Serving Layer

- Designed and integrated a Netflix DGS subgraph into CarGurus' federated GraphQL gateway, creating a unified API layer for frontend, backend, and AI consumers from a single graph with consistent contracts.
- Implemented schema federation, batch data loaders (N+1 prevention), and JVM warmup strategies; built load tests and integration tests ensuring production-grade reliability and API quality.

Digital Commerce Platform & Experimentation

- Led backend design and launch of multiple new products using microservice and event-driven architecture; enabled the company to quickly launch and safely retire product experiments without destabilizing core systems.

Cross-Org Platform Leadership & Engineering Velocity

- Established patterns for API design, caching, and distributed orchestration adopted across multiple teams; authored key design docs, RFCs, and decision records that drove consensus across platform and product teams.
- Mentored multiple senior engineers on system design, API craft, and path to Principal-level scope.

Applause — Senior Software Engineer / Technical Lead

Oct 2017 – Oct 2019

Authentication & Identity Platform — Enterprise Security

- Upgraded legacy authentication to a modern Keycloak-based SSO platform, unifying auth across internal systems, an enterprise partner platform, and a large external community. Introduced social and mobile login; migrated multiple Angular, Node.js, and Java monoliths behind the new auth gateway with minimal downtime.

Event Sourcing & CQRS Platform Initiative

- Drove design and prototyping of a microservice architecture based on event sourcing and CQRS; defined foundational patterns for services, events, and read/write models adopted by new domains.

Cross-Org Collaboration & Enterprise Integrations

- Partnered with product, support, and customer-facing teams to modernize internal workflows and enterprise integrations with clear observability, rollback paths, and compliance considerations.

Toast Inc. — Senior Software Engineer (Early Engineer)

Sep 2014 – Oct 2017

Monolith → Microservices — Founding Platform Services

- Among the earliest engineers during hypergrowth (first customer to hundreds in a few years); designed and implemented core backend services (Order Service, Labor Service, foundational APIs) that became the basis for the platform's growth to billions in GMV.
- Built initial AWS deployment pipelines before a dedicated DevOps team existed; created the first automated testing framework, enabling safer releases during rapid product and team expansion.

Third-Party API Integrations & Partner Platform

- Led integrations with key partners (e.g., workforce management), designing reliable bidirectional sync and schema evolution. Partnered with PMs, sales, and support to translate customer feedback into API and platform capabilities.

The MathWorks Inc. — Software Engineer II

Jul 2013 – Aug 2014

Developed internal engineering tools for MATLAB/Simulink workflows. Improved backend automation systems for build/test reliability and developer productivity.

EDUCATION

M.S. in Electrical and Computer Engineering — Northeastern University, Boston 2010 – 2013

B.S. in Information Science — Beijing University of Aeronautics and Astronautics 2006 – 2010